

MAT 3150 — Homework 2

Due September 17

Problem 1 (10 points)

Prove that if x and y are arbitrary real numbers such that $x < y$, there exists at least one irrational number z satisfying $x < z < y$.

Proof. By the density of $\mathbb{Q}$ in $\mathbb{R}$, we can find a rational number r_1 such that $x < r_1 < y$. Applying density again to the interval (r_1, y) , we can find a rational number r_2 such that $r_1 < r_2 < y$.

Now consider

$$z = r_1 + \frac{r_2 - r_1}{\sqrt{2}}.$$

z is **irrational**. Suppose, for the sake of contradiction, that z is rational. Then

$$z - r_1 = \frac{r_2 - r_1}{\sqrt{2}}$$

has rational left-hand side (since z and r_1 are both rational). So $\frac{r_2 - r_1}{\sqrt{2}}$ is rational, say equal to q . Since $r_2 - r_1 \neq 0$ (as $r_1 < r_2$), we get $\sqrt{2} = \frac{r_2 - r_1}{q}$, a ratio of two rationals, so $\sqrt{2}$ would be rational. This contradicts the irrationality of $\sqrt{2}$. Hence z is irrational.

$x < z < y$. Since $\sqrt{2} > 1$ and $r_2 - r_1 > 0$,

$$0 < \frac{r_2 - r_1}{\sqrt{2}} < r_2 - r_1.$$

Adding r_1 throughout gives $r_1 < z < r_2$. Combined with $x < r_1$ and $r_2 < y$, this yields

$$x < r_1 < z < r_2 < y,$$

so in particular $x < z < y$. □

Problem 2 (10 points)

Find the supremum and the infimum of the sets

$$S_1 = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\} \quad \text{and} \quad S_2 = \left\{ \frac{(-1)^n}{n} : n \in \mathbb{N} \right\}.$$

S_1

By using the Archimedean property, we can show that $\inf S_1 = 0$. For any $\epsilon > 0$, there exists $n \in \mathbb{N}$ such that $1/n < \epsilon$, which means that 0 is indeed the greatest lower bound of S_1 . The supremum of S_1 is $\sup S_1 = 1$, since the largest element of the set is 1 (attained when $n = 1$).

S_2

For even n the term $(-1)^n/n = 1/n$ is positive, and for odd n the term is $-1/n$, negative. So the candidate supremum comes from the even terms and the candidate infimum from the odd terms.

$\sup S_2 = \frac{1}{2}$, **attained at** $n = 2$. Among the even terms $1/n$, the largest occurs at the smallest even n , namely $n = 2$, giving $1/2$; every other even term satisfies $1/n \leq 1/2$ for $n \geq 2$. Every odd term is negative, hence $< 1/2$. So $1/2$ is an upper bound for S_2 , and since $1/2 \in S_2$ it is the maximum, hence $\sup S_2 = 1/2$.

$\inf S_2 = -1$, **attained at** $n = 1$. Among the odd terms $-1/n$, the smallest (most negative) occurs at the smallest odd n , namely $n = 1$, giving -1 ; every other odd term satisfies $-1/n \geq -1$ for $n \geq 1$. Every even term is positive, hence > -1 . So -1 is a lower bound for S_2 , and since $-1 \in S_2$ it is the minimum, hence $\inf S_2 = -1$.

Problem 3 (10 points)

If T, S are nonempty sets so that $s \leq t$ for all $s \in S$ and $t \in T$, prove that $\sup S$ and $\inf T$ exist, and prove the inequality $\sup S \leq \inf T$.

Proof. Step 1: $\sup S$ exists. S is nonempty by hypothesis. Since T is nonempty, fix some $t_0 \in T$. By hypothesis $s \leq t_0$ for every $s \in S$, so t_0 is an upper bound for S . Thus S is nonempty and bounded above, so by the completeness axiom, $\sup S$ exists.

Step 2: $\inf T$ exists. T is nonempty by hypothesis. Since S is nonempty, fix some $s_0 \in S$. By hypothesis $s_0 \leq t$ for every $t \in T$, so s_0 is a lower bound for T . Thus T is nonempty and bounded below, so by the completeness axiom, $\inf T$ exists.

Step 3: $\sup S \leq \inf T$. Fix an arbitrary $t \in T$. Since $s \leq t$ for every $s \in S$, t is an upper bound for S , so by the least upper bound property $\sup S \leq t$. This holds for every $t \in T$, so $\sup S$ is a lower bound for T . Since $\inf T$ is the greatest lower bound of T , it follows that $\sup S \leq \inf T$. $\square$

Problem 4 (10 points)

For A and B nonempty bounded subsets of real numbers, let $A + B$ be the set of all sums $a + b$, where $a \in A$ and $b \in B$. Prove that

$$\sup(A + B) = \sup A + \sup B.$$

Proof. Let $\alpha = \sup A$ and $\beta = \sup B$; these exist by the completeness axiom since A and B are nonempty and bounded above.

Step 1: $\alpha + \beta$ is an upper bound for $A + B$. Let $a \in A$ and $b \in B$ be arbitrary. Since $\alpha = \sup A$, $a \leq \alpha$; since $\beta = \sup B$, $b \leq \beta$. Adding, $a + b \leq \alpha + \beta$. As a, b were arbitrary, every element of $A + B$ is $\leq \alpha + \beta$, so $\alpha + \beta$ is an upper bound for $A + B$. In particular, $A + B$ is nonempty and bounded above, so $\sup(A + B)$ exists and

$$\sup(A + B) \leq \alpha + \beta.$$

Step 2: $\alpha + \beta$ is the least upper bound. Let $\varepsilon > 0$ be arbitrary. Since $\alpha = \sup A$, $\alpha - \varepsilon/2$ is not an upper bound for A , so there exists $a \in A$ with $a > \alpha - \varepsilon/2$. Similarly, since $\beta = \sup B$, there exists $b \in B$ with $b > \beta - \varepsilon/2$. Then $a + b \in A + B$ and

$$a + b > \left(\alpha - \frac{\varepsilon}{2}\right) + \left(\beta - \frac{\varepsilon}{2}\right) = \alpha + \beta - \varepsilon.$$

Thus for every $\varepsilon > 0$ there is an element of $A + B$ exceeding $\alpha + \beta - \varepsilon$, so no number strictly less than $\alpha + \beta$ can be an upper bound for $A + B$. Hence

$$\sup(A + B) \geq \alpha + \beta.$$

Combining Steps 1 and 2, $\sup(A + B) = \alpha + \beta = \sup A + \sup B$.

□